

DMD Project

1. The motivation behind DMD

Dynamic Mode Decomposition(DMD) is motivated by the need to analyze high-dimensional time-dependent data in a compact way. In many dynamical systems, the essential behavior is often governed by only a few dominant patterns. The observed data may appear complicated, so we need to find a way to extract meaningful dynamical structures effectively. DMD addresses this problem by approximating the evolution between consecutive snapshots with a linear operator. Based on this approximation, it identifies dynamic modes and eigenvalues that can be used to reconstruct observed data and make short-term predictions. From the Koopman viewpoint, DMD can even provide meaningful spectral information for certain nonlinear systems.

2. The Basic Mathematical Setting

Suppose we observe a dynamical system at m consecutive time steps and collect the snapshots

$$x_1, x_2, \dots, x_m \in \mathbb{R}^n$$

Each snapshot x_k is a high-dimensional state vector. To describe the one-step evolution of the system, we form two data matrices

$$X = [x_1, \dots, x_{m-1}], \quad X' = [x_2, \dots, x_m]$$

The main idea of DMD is to assume that the evolution from one snapshot to the next can be approximated by a linear operator A , so that

$$X' \approx AX$$

Since the original dimension n can be very large, we apply truncated SVD to X :

$$X \approx U_r \Sigma_r V_r^*$$

This gives a lower-dimensional representation of the data. Here, U_r contains the dominant spatial structures, Σ_r contains the leading singular values, and V_r contains the corresponding temporal information.

3. The Exact DMD Idea

From the approximation $X' \approx AX$, we write the least-squares evolution operator as

$$A \approx X'X^\dagger$$

Using the truncated SVD of X , we obtain

$$X^\dagger = V_r \Sigma_r^{-1} U_r^*$$

so

$$A \approx X'V_r \Sigma_r^{-1} U_r^*$$

Instead of working with the large matrix A , exact DMD uses the reduced operator

$$A_r = U_r^* X' V_r \Sigma_r^{-1}$$

Then we compute

$$A_r W = W \Lambda$$

where Λ contains the DMD eigenvalues. These eigenvalues describe the temporal behavior of the system, such as oscillation, growth, and decay.

The exact DMD modes are defined by

$$\Phi = X' V_r \Sigma_r^{-1} W$$

This is different from the projected form $U_r W$. The matrix $U_r W$ only gives modes in the reduced subspace, while Φ gives modes in the original state space. Therefore, exact DMD provides a more accurate description of the full dynamics.

After that, the initial amplitudes are computed by

$$b = \Phi^\dagger x_1$$

The DMD model is then written as

$$x_k = \Phi \Lambda^{k-1} b$$

When $k \leq m$, this formula is used for reconstruction. When $k > m$, it can be used for short-term prediction.

4. Extensions of DMD: Time-Delay DMD

Exact DMD uses the relation between consecutive snapshots and assumes that the dynamics can be approximated by a single linear map. However, for more complicated systems, one snapshot may not contain enough information about the temporal evolution. Time-delay DMD addresses this by stacking several consecutive snapshots into a delay vector

$$y_k = \begin{bmatrix} x_k \\ x_{k+1} \\ \vdots \\ x_{k+d-1} \end{bmatrix}$$

DMD is then applied to these delay-embedded states. This gives the model short temporal memory and may help capture richer dynamics than exact DMD. In this project, time-delay DMD is used as the extension method and compared with exact DMD on the main spatio-temporal dataset.

From the Koopman viewpoint, DMD approximates the evolution of observables by a finite-dimensional linear operator. Time-Delay DMD augments the observables with delayed snapshots, providing a richer representation of the dynamics. This often leads to a better approximation for systems with more complex temporal structure or weak nonlinearity.

5. Data Generation and Error Metrics

5.1 Data Generation

Two synthetic datasets were used in this project. The first one is a simple linear oscillator with easy-to-interpret behavior. The second one is the main spatio-temporal dataset used for comparing the correctness of exact DMD and time-delay DMD.

For the linear validation dataset, the state evolves according to

$$x_{k+1} = A_{\text{true}} x_k$$

Where

$$A_{\text{true}} = \begin{bmatrix} e^{\alpha \Delta t} \cos(\omega \Delta t) & -e^{\alpha \Delta t} \sin(\omega \Delta t) \\ e^{\alpha \Delta t} \sin(\omega \Delta t) & e^{\alpha \Delta t} \cos(\omega \Delta t) \end{bmatrix}$$

For the spatio-temporal dataset, I generate a spatio-temporal field by combining two localized modes with different temporal frequencies:

$$f_1(x, t) = \frac{1}{\cosh(x + 2.5)} \cos(2.2t)$$

$$f_2(x, t) = \frac{1.6}{\cosh(x - 0.5)} \tanh(x - 0.5) \sin(2.8t)$$

The full dataset is defined by

$$X(x, t) = f_1(x, t) + f_2(x, t) + \varepsilon(x, t)$$

Where $\varepsilon(x, t)$ is a small noise term.

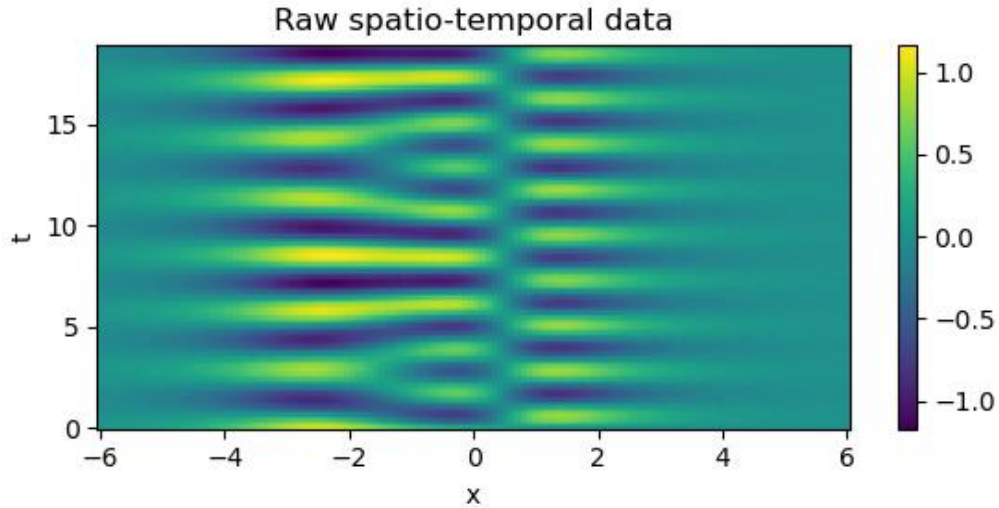


Figure 1. Raw spatio-temporal dataset

5.2 Error Metrics

To evaluate the reconstruction and prediction performance, I used the relative Frobenius norm error

$$\frac{\|X - X_{\text{DMD}}\|_F}{\|X\|_F}$$

Here, X denotes the true data matrix and X_{DMD} denotes the reconstructed or predicted data matrix. A small

6. Numerical Experiments

6.1 Exact DMD on a Linear Validation Dataset

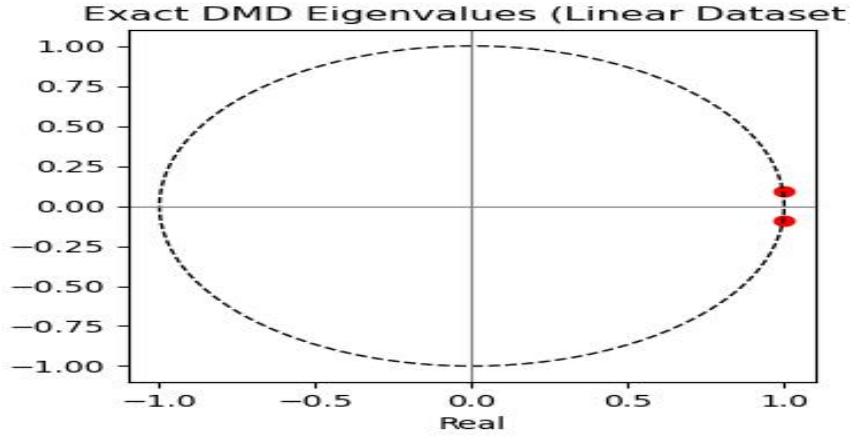


Figure 2. Exact DMD eigenvalues (Linear Dataset)

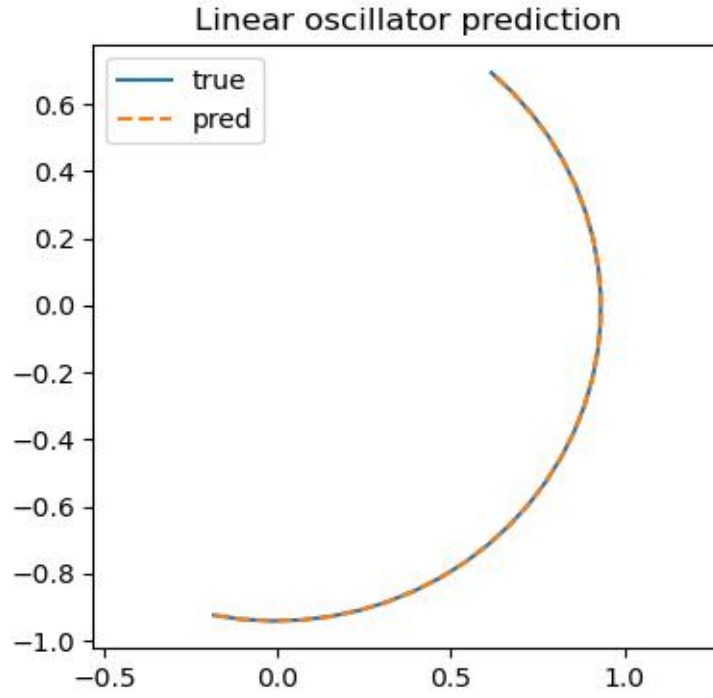


Figure 3. Linear oscillator prediction

I first tested exact DMD on the linear validation dataset. Figure 2 shows that the recovered eigenvalues form a complex-conjugate pair near the unit circle, which is consistent with oscillatory dynamics since the two points lie close to the unit circle. This indicates that exact DMD captures the main spectral behavior of linear system.

Figure 3 shows the short-term prediction result. The predicted trajectory is almost identical to the true trajectory, which confirms that the exact DMD implementation works well on simple low-dimensional linear dynamics.

Support data: Linear reconstruction error: 1.0381690838463442e-14

6.2 Exact DMD and Time-Delay DMD on the Spatio-Temporal Dataset

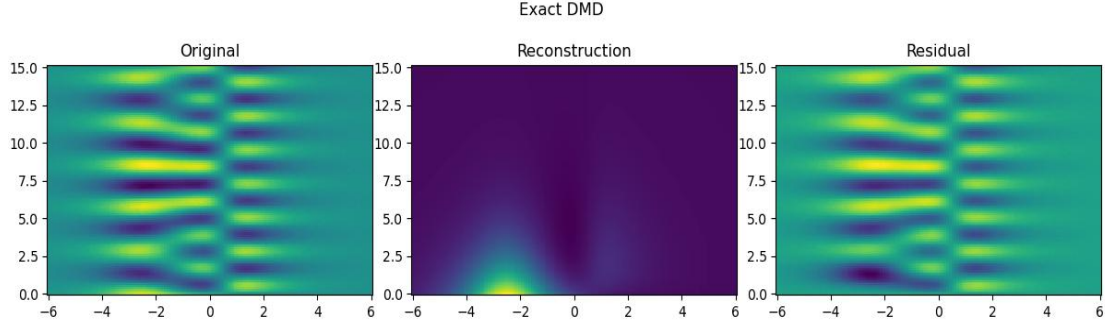


Figure 4. Exact DMD reconstruction

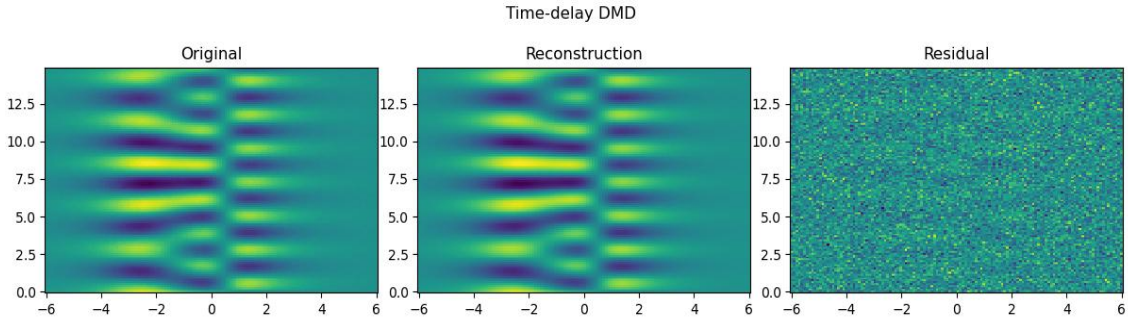


Figure 5. Time-delay DMD reconstruction

I then applied both exact DMD and time-delay DMD to the main spatio-temporal dataset.

The reconstruction result of exact DMD is shown in Figure 4. Although the method captures part of the global structure, the residual remains large, which indicates that exact DMD is not sufficient for this more complicated dataset.

By contrast, the time-delay DMD result in Figure 5 is much closer to the original field. The residual is substantially smaller, which suggests that delay embedding helps capture additional temporal structure.

Therefore, in this spatio-temporal dataset, time-delay DMD performs much better.

Support data: Spatio-temporal exact rec error: 1.0465056766359349
 Spatio-temporal exact pred error: 1.0000145575861825
 Spatio-temporal delay rec error: 0.026738531909023738
 Spatio-temporal delay pred error: 0.5855335056974422

6.3 Effect of Rank Truncation

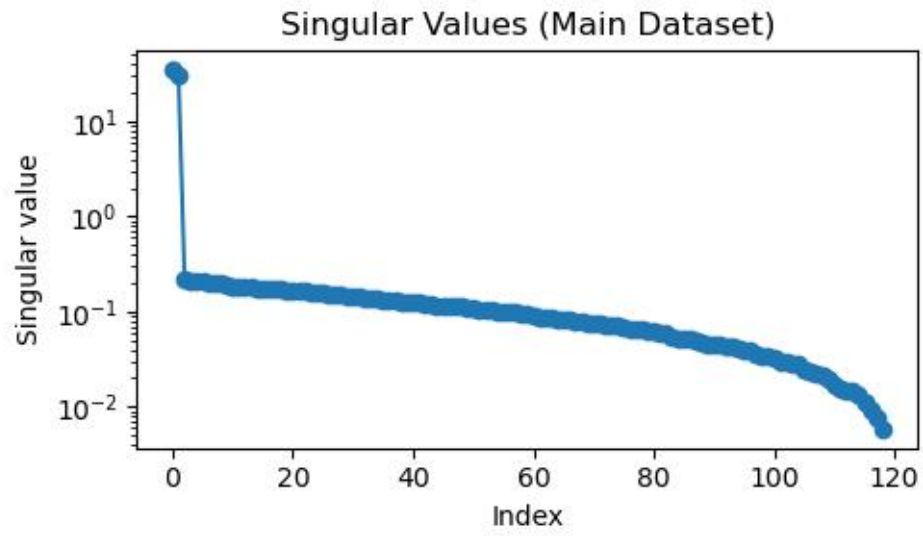


Figure 6. Singular values (Main Dataset)

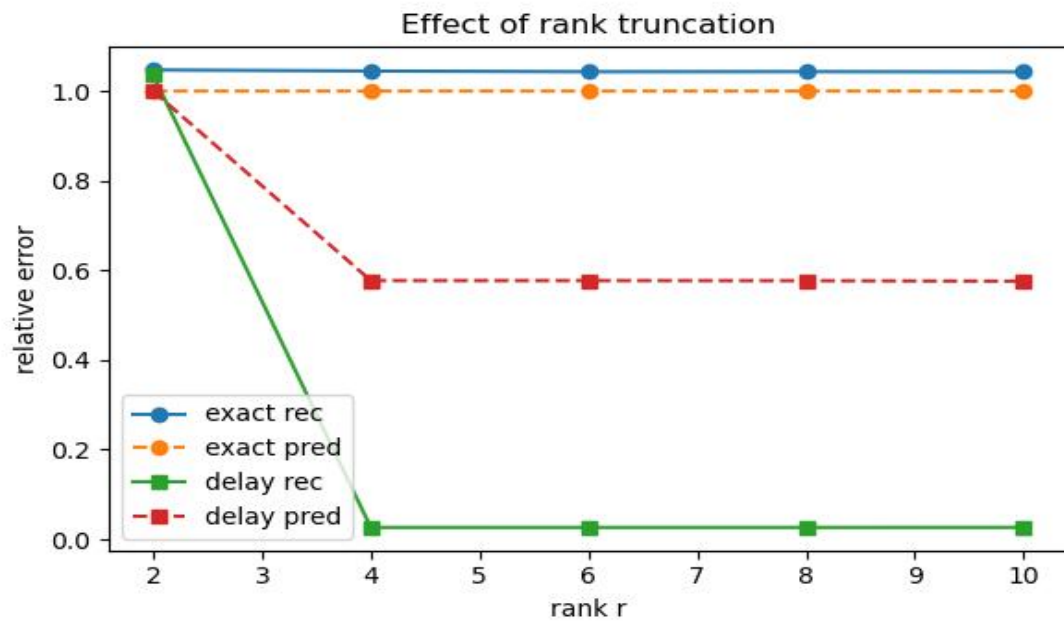


Figure 7. Effect of rank truncation

Before comparing the effect of different truncation ranks, we first examine the singular value decay of the main dataset. As shown in Figure 6, there is one dominant singular value, but the remaining singular values do not drop to zero immediately. This suggests that the dataset contains richer structures may not be captured by an extreme low-rank truncation. Therefore, it is meaningful to study how the truncation rank r affects the performance of DMD.

Figure 7 compares the reconstruction and prediction errors of exact DMD and time-delay DMD for different values of r . For exact DMD, the errors remain relatively high across the tested ranks, which suggests that changing the rank alone does not resolve the model mismatch. For time-delay DMD, the reconstruction error decreases sharply to a very low value when r is sufficiently large, which means using $r = 4$ is sufficient for reconstruction in this setting. But for prediction, the error remains large though it is much better than the prediction error of exact DMD. Accurate prediction on the main dataset remains difficult. Time-delay DMD can represent the observed data well, but future extrapolation is more sensitive to errors in eigenvalues, modal amplitudes, noise, and model mismatch.

7. Conclusion

In this project, I implemented exact DMD and time-delay DMD on two synthetic datasets. Exact DMD worked well on the simple linear validation dataset and produced accurate short-term prediction. On the main spatio-temporal dataset, exact DMD gave poor reconstruction, while time-delay DMD significantly improved the reconstruction by using delayed snapshots. However, prediction on the main dataset remained challenging, partly because the data contain richer temporal structures and a small noise term. In conclusion, exact DMD is suitable for simple linear dynamics, while time-delay DMD is more effective for reconstructing more complicated spatio-temporal data.

The full Python code is attached as Project 2 code.py.

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